# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 83.7 ± 1.19 |
| Correlation Stability | 60.9 ± 2.02 |
| Distribution Consistency | 89.7 ± 0.64 |
| Entropy Preservation | 90.1 ± 1.04 |
| Mean Z Score Drift | 0.2 ± 0.02 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 61.1 ± 0.30 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 29.1 ± 2.39 |
| Feature Accuracy | 87.1 ± 1.64 |
| Novelty Score | 0.3 ± 0.01 |

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## Dataset Audit: indian_diabetes.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 768 |
| Columns | 9 |
| Numeric features | 9 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4o-mini |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| Pregnancies | 3.845 | 3.282 | -0.563 | 14.6% | Warn |
| Glucose | 120.895 | 130.394 | +9.499 | 7.9% | OK |
| BloodPressure | 69.105 | 74.687 | +5.581 | 8.1% | OK |
| SkinThickness | 20.536 | 27.036 | +6.500 | 31.6% | Drift |
| Insulin | 79.799 | 75.339 | -4.461 | 5.6% | OK |
| BMI | 31.993 | 32.322 | +0.330 | 1.0% | OK |
| DiabetesPedigreeFunction | 0.472 | 0.354 | -0.118 | 24.9% | Warn |
| Age | 33.241 | 32.408 | -0.833 | 2.5% | OK |
| Outcome | 0.349 | 0.483 | +0.135 | 38.5% | Drift |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 83.7 ± 1.19 |
| Correlation Stability (*2) | 60.9 ± 2.02 |
| Distribution Consistency (*3) | 89.7 ± 0.64 |
| Entropy Preservation (*4) | 90.1 ± 1.04 |
| Mean Z Score Drift (*5) | 0.2 ± 0.02 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 61.1 ± 0.30 |
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 29.1 ± 2.39 |
| Feature Accuracy (*10) | 87.1 ± 1.64 |

| | |
|---|---|
| Novelty Score (*11) | 0.3 ± 0.01 |

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7727272727272727 | 0.6512987012987013 | 0.7487012987012986 | -0.02402597402597405 | |
| F1 | 0.6278681287466419 | 0.5761951416763572 | 0.6444970375600928 | 0.01662890881345004 8 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6941558441558442 | 0.6422077922077922 | 0.6824675324675324 | -0.01168831168831169 2 | |
| F1 | 0.5116012374545862 | 0.5696266048101221 | 0.5442545175802431 | 0.03265328012565697 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7493506493506492 | 0.7084415584415584 | 0.7642857142857143 | 0.01493506493506495 7 | |
| F1 | 0.6076604413342895 | 0.5978426637741581 | 0.6480884247837252 | 0.04042798344943551 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7545454545454545 | 0.7090909090909091 | 0.7480519480519481 | -0.00649350649350649 6 | |
| F1 | 0.5998259463389416 | 0.5694580257519984 | 0.6350788570769061 | 0.03525291073796453 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7506493506493507 | 0.7188311688311689 | 0.7564935064935064 | 0.00584415584415583 | |
| F1 | 0.6051923812479628 | 0.6260776666978077 | 0.6154537157829814 | 0.01026133453501850 7 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7636363636363636 | 0.659090909090909 | 0.7435064935064934 | -0.02012987012987014 | |
| F1 | 0.6399852354635025 | 0.6095004292936327 | 0.6417035018112817 | 0.00171826634777928 47 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7064935064935065 | 0.6318181818181818 | 0.6928571428571428 | -0.01363636363636364 6 | |
| F1 | 0.5663351640264895 | 0.5394516387153327 | 0.5623273829819443 | -0.00400778104454530 5 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7564935064935064 | 0.7168831168831169 | 0.7655844155844156 | 0.00909090909090907 1 | |
| F1 | 0.5733680734057942 | 0.5800666066979847 | 0.6375579669455058 | 0.06418989353971156 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7272727272727273 | 0.7077922077922079 | 0.7259740259740260 | -0.00129870129870131 02 | |
| F1 | 0.5906687746013825 | 0.6020857564041895 | 0.6041789868208257 | 0.01351021221944321 2 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7448051948051948 | 0.6987012987012987 | 0.7435064935064936 | -0.00129870129870132 13 | |
| F1 | 0.6134889060283798 | 0.6038192196284434 | 0.6259432177912763 | 0.01245431176289644 1 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunct | Age | Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.0 | 148.0 | 72.0 | 35.0 | 0.0 | 33.6 | 0.627 | 50.0 | 1.0 |
| 2 | 1.0 | 85.0 | 66.0 | 29.0 | 0.0 | 26.6 | 0.351 | 31.0 | 0.0 |
| 3 | 8.0 | 183.0 | 64.0 | 0.0 | 0.0 | 23.3 | 0.672 | 32.0 | 1.0 |
| 4 | 1.0 | 89.0 | 66.0 | 23.0 | 94.0 | 28.1 | 0.167 | 21.0 | 0.0 |
| 5 | 0.0 | 137.0 | 40.0 | 35.0 | 168.0 | 43.1 | 2.288 | 33.0 | 1.0 |
| 6 | 5.0 | 116.0 | 74.0 | 0.0 | 0.0 | 25.6 | 0.201 | 30.0 | 0.0 |
| 7 | 3.0 | 78.0 | 50.0 | 32.0 | 88.0 | 31.0 | 0.248 | 26.0 | 1.0 |
| 8 | 10.0 | 115.0 | 0.0 | 0.0 | 0.0 | 35.3 | 0.134 | 29.0 | 0.0 |
| 9 | 2.0 | 197.0 | 70.0 | 45.0 | 543.0 | 30.5 | 0.158 | 53.0 | 1.0 |
| 10 | 8.0 | 125.0 | 96.0 | 0.0 | 0.0 | 0.0 | 0.232 | 54.0 | 1.0 |
| 11 | 4.0 | 110.0 | 92.0 | 0.0 | 0.0 | 37.6 | 0.191 | 30.0 | 0.0 |
| 12 | 10.0 | 168.0 | 74.0 | 0.0 | 0.0 | 38.0 | 0.537 | 34.0 | 1.0 |
| 13 | 10.0 | 139.0 | 80.0 | 0.0 | 0.0 | 27.1 | 1.441 | 57.0 | 0.0 |
| 14 | 1.0 | 189.0 | 60.0 | 23.0 | 846.0 | 30.1 | 0.398 | 59.0 | 1.0 |
| 15 | 5.0 | 166.0 | 72.0 | 19.0 | 175.0 | 25.8 | 0.587 | 51.0 | 1.0 |
| 16 | 7.0 | 100.0 | 0.0 | 0.0 | 0.0 | 30.0 | 0.484 | 32.0 | 1.0 |
| 17 | 0.0 | 118.0 | 84.0 | 47.0 | 230.0 | 45.8 | 0.551 | 31.0 | 1.0 |
| 18 | 7.0 | 107.0 | 74.0 | 0.0 | 0.0 | 29.6 | 0.254 | 31.0 | 1.0 |
| 19 | 1.0 | 103.0 | 30.0 | 38.0 | 83.0 | 43.3 | 0.183 | 33.0 | 0.0 |
| 20 | 1.0 | 115.0 | 70.0 | 30.0 | 96.0 | 34.6 | 0.529 | 32.0 | 1.0 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 99.0 | 0.0 | 0.0 | 0.0 | 25.0 | 0.253 | 22.0 | 0.0 | 1.0 |
| 2 | 1.0 | 88.0 | 30.0 | 42.0 | 99.0 | 55.0 | 0.496 | 26.0 | 1.0 | 1.0 |
| 3 | 2.0 | 85.0 | 65.0 | 0.0 | 0.0 | 39.6 | 0.93 | 27.0 | 0.0 | 1.0 |
| 4 | 5.0 | 122.0 | 86.0 | 0.0 | 0.0 | 34.7 | 0.29 | 33.0 | 0.0 | 1.0 |
| 5 | 1.0 | 90.0 | 68.0 | 8.0 | 0.0 | 24.5 | 1.138 | 36.0 | 0.0 | 1.0 |
| 6 | 2.0 | 81.0 | 72.0 | 15.0 | 76.0 | 30.1 | 0.547 | 25.0 | 0.0 | 1.0 |
| 7 | 3.0 | 145.0 | 70.0 | 35.0 | 0.0 | 32.1 | 0.627 | 29.0 | 1.0 | 1.0 |
| 8 | 1.0 | 85.0 | 60.0 | 20.0 | 0.0 | 24.5 | 0.351 | 22.0 | 0.0 | 1.0 |
| 9 | 0.0 | 120.0 | 80.0 | 25.0 | 90.0 | 36.2 | 0.456 | 27.0 | 1.0 | 1.0 |
| 10 | 4.0 | 110.0 | 68.0 | 30.0 | 0.0 | 28.5 | 0.234 | 34.0 | 0.0 | 1.0 |
| 11 | 2.0 | 130.0 | 72.0 | 28.0 | 150.0 | 31.4 | 0.512 | 25.0 | 1.0 | 1.0 |
| 12 | 5.0 | 140.0 | 75.0 | 20.0 | 0.0 | 29.8 | 0.389 | 38.0 | 0.0 | 1.0 |
| 13 | 8.0 | 110.0 | 76.0 | 0.0 | 0.0 | 27.8 | 0.237 | 58.0 | 0.0 | nan |
| 14 | 1.0 | 90.0 | 62.0 | 18.0 | 59.0 | 25.1 | 1.268 | 25.0 | 0.0 | nan |
| 15 | 2.0 | 56.0 | 56.0 | 28.0 | 45.0 | 24.2 | 0.332 | 22.0 | 0.0 | nan |
| 16 | 5.0 | 111.0 | 72.0 | 28.0 | 0.0 | 23.9 | 0.407 | 27.0 | 0.0 | nan |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.0 | 126.0 | 78.0 | 27.0 | 22.0 | 29.6 | 0.439 | 40.0 | 0.0 | 2.0 |
| 2 | 4.0 | 99.0 | 72.0 | 17.0 | 0.0 | 25.6 | 0.294 | 28.0 | 0.0 | 2.0 |
| 3 | 4.0 | 115.0 | 72.0 | 0.0 | 0.0 | 28.9 | 0.376 | 46.0 | 1.0 | 2.0 |
| 4 | 2.0 | 175.0 | 88.0 | 0.0 | 0.0 | 22.9 | 0.326 | 22.0 | 0.0 | 2.0 |
| 5 | 5.0 | 88.0 | 78.0 | 30.0 | 0.0 | 27.6 | 0.258 | 37.0 | 0.0 | 2.0 |
| 6 | 1.0 | 128.0 | 88.0 | 39.0 | 110.0 | 36.5 | 1.057 | 37.0 | 1.0 | 2.0 |
| 7 | 3.0 | 135.0 | 70.0 | 20.0 | 0.0 | 32.5 | 0.345 | 30.0 | 1.0 | 2.0 |
| 8 | 6.0 | 150.0 | 80.0 | 25.0 | 150.0 | 36.2 | 0.456 | 45.0 | 1.0 | 2.0 |
| 9 | 1.0 | 90.0 | 60.0 | 15.0 | 0.0 | 28.4 | 0.234 | 22.0 | 0.0 | 2.0 |
| 10 | 4.0 | 110.0 | 72.0 | 30.0 | 80.0 | 30.1 | 0.289 | 35.0 | 0.0 | 2.0 |
| 11 | 5.0 | 145.0 | 85.0 | 35.0 | 200.0 | 37.5 | 0.567 | 50.0 | 1.0 | 2.0 |
| 12 | 2.0 | 120.0 | 65.0 | 10.0 | 0.0 | 26.8 | 0.189 | 28.0 | 0.0 | 2.0 |
| 13 | 0.0 | 91.0 | 68.0 | 32.0 | 210.0 | 39.9 | 0.381 | 25.0 | 0.0 | nan |
| 14 | 3.0 | 173.0 | 82.0 | 48.0 | 465.0 | 38.4 | 2.137 | 25.0 | 1.0 | nan |
| 15 | 1.0 | 147.0 | 94.0 | 41.0 | 0.0 | 49.3 | 0.358 | 27.0 | 1.0 | nan |
| 16 | 2.0 | 107.0 | 74.0 | 30.0 | 100.0 | 33.6 | 0.404 | 23.0 | 0.0 | nan |

## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 13.0 | 106.0 | 70.0 | 0.0 | 0.0 | 34.2 | 0.251 | 52.0 | 0.0 | 3.0 |
| 2 | 8.0 | 109.0 | 76.0 | 39.0 | 114.0 | 27.9 | 0.64 | 31.0 | 1.0 | 3.0 |
| 3 | 5.0 | 136.0 | 84.0 | 41.0 | 88.0 | 35.0 | 0.286 | 35.0 | 1.0 | 3.0 |
| 4 | 15.0 | 136.0 | 70.0 | 32.0 | 110.0 | 37.1 | 0.153 | 43.0 | 1.0 | 3.0 |
| 5 | 0.0 | 108.0 | 68.0 | 20.0 | 0.0 | 27.3 | 0.787 | 32.0 | 0.0 | 3.0 |
| 6 | 13.0 | 152.0 | 90.0 | 33.0 | 29.0 | 26.8 | 0.731 | 43.0 | 1.0 | 3.0 |
| 7 | 3.0 | 145.0 | 70.0 | 25.0 | 80.0 | 32.5 | 0.567 | 29.0 | 1.0 | 3.0 |
| 8 | 1.0 | 90.0 | 60.0 | 20.0 | 0.0 | 24.5 | 0.123 | 22.0 | 0.0 | 3.0 |
| 9 | 5.0 | 160.0 | 85.0 | 30.0 | 150.0 | 36.2 | 0.456 | 34.0 | 1.0 | 3.0 |
| 10 | 0.0 | 110.0 | 65.0 | 15.0 | 0.0 | 22.1 | 0.321 | 27.0 | 0.0 | 3.0 |
| 11 | 4.0 | 130.0 | 75.0 | 28.0 | 200.0 | 34.0 | 0.678 | 40.0 | 1.0 | 3.0 |
| 12 | 2.0 | 95.0 | 68.0 | 22.0 | 50.0 | 26.5 | 0.234 | 23.0 | 0.0 | 3.0 |
| 13 | 5.0 | 158.0 | 84.0 | 41.0 | 210.0 | 39.4 | 0.395 | 29.0 | 1.0 | nan |
| 14 | 1.0 | 101.0 | 50.0 | 15.0 | 36.0 | 24.2 | 0.526 | 26.0 | 0.0 | nan |
| 15 | 0.0 | 109.0 | 88.0 | 30.0 | 0.0 | 32.5 | 0.855 | 38.0 | 1.0 | nan |
| 16 | 5.0 | 147.0 | 75.0 | 0.0 | 0.0 | 29.9 | 0.434 | 28.0 | 0.0 | nan |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.0 | 100.0 | 66.0 | 15.0 | 56.0 | 23.6 | 0.666 | 26.0 | 0.0 | 4.0 |
| 2 | 2.0 | 101.0 | 58.0 | 17.0 | 265.0 | 24.2 | 0.614 | 23.0 | 0.0 | 4.0 |
| 3 | 5.0 | 147.0 | 78.0 | 0.0 | 0.0 | 33.7 | 0.218 | 65.0 | 0.0 | 4.0 |
| 4 | 9.0 | 145.0 | 80.0 | 46.0 | 130.0 | 37.9 | 0.637 | 40.0 | 1.0 | 4.0 |
| 5 | 4.0 | 97.0 | 60.0 | 23.0 | 0.0 | 28.2 | 0.443 | 22.0 | 0.0 | 4.0 |
| 6 | 5.0 | 162.0 | 104.0 | 0.0 | 0.0 | 37.7 | 0.151 | 52.0 | 1.0 | 4.0 |
| 7 | 3.0 | 145.0 | 72.0 | 25.0 | 150.0 | 34.5 | 0.567 | 29.0 | 1.0 | 4.0 |
| 8 | 1.0 | 85.0 | 60.0 | 20.0 | 0.0 | 22.1 | 0.123 | 24.0 | 0.0 | 4.0 |
| 9 | 5.0 | 160.0 | 80.0 | 30.0 | 200.0 | 36.2 | 0.456 | 34.0 | 1.0 | 4.0 |
| 10 | 2.0 | 110.0 | 70.0 | 15.0 | 80.0 | 28.9 | 0.234 | 27.0 | 0.0 | 4.0 |
| 11 | 4.0 | 130.0 | 75.0 | 35.0 | 120.0 | 31.5 | 0.345 | 30.0 | 1.0 | 4.0 |
| 12 | 0.0 | 95.0 | 65.0 | 10.0 | 0.0 | 24.3 | 0.198 | 22.0 | 0.0 | 4.0 |
| 13 | 5.0 | 117.0 | 86.0 | 30.0 | 105.0 | 39.1 | 0.251 | 42.0 | 0.0 | nan |
| 14 | 3.0 | 74.0 | 68.0 | 28.0 | 45.0 | 29.7 | 0.293 | 23.0 | 0.0 | nan |
| 15 | 2.0 | 129.0 | 0.0 | 0.0 | 0.0 | 38.5 | 0.304 | 41.0 | 0.0 | nan |
| 16 | 1.0 | 126.0 | 56.0 | 29.0 | 152.0 | 28.7 | 0.801 | 21.0 | 0.0 | nan |

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## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4.0 | 99.0 | 68.0 | 38.0 | 0.0 | 32.8 | 0.145 | 33.0 | 0.0 | 5.0 |
| 2 | 0.0 | 102.0 | 75.0 | 23.0 | 0.0 | 0.0 | 0.572 | 21.0 | 0.0 | 5.0 |
| 3 | 2.0 | 92.0 | 62.0 | 28.0 | 0.0 | 31.6 | 0.13 | 24.0 | 0.0 | 5.0 |
| 4 | 10.0 | 122.0 | 78.0 | 31.0 | 0.0 | 27.6 | 0.512 | 45.0 | 0.0 | 5.0 |
| 5 | 4.0 | 110.0 | 76.0 | 20.0 | 100.0 | 28.4 | 0.118 | 27.0 | 0.0 | 5.0 |
| 6 | 0.0 | 189.0 | 104.0 | 25.0 | 0.0 | 34.3 | 0.435 | 41.0 | 1.0 | 5.0 |
| 7 | 3.0 | 85.0 | 70.0 | 22.0 | 0.0 | 28.5 | 0.4 | 30.0 | 0.0 | 5.0 |
| 8 | 1.0 | 150.0 | 80.0 | 15.0 | 0.0 | 31.2 | 0.5 | 24.0 | 0.0 | 5.0 |
| 9 | 5.0 | 120.0 | 78.0 | 30.0 | 100.0 | 34.5 | 0.6 | 35.0 | 1.0 | 5.0 |
| 10 | 2.0 | 95.0 | 64.0 | 20.0 | 0.0 | 26.8 | 0.3 | 28.0 | 0.0 | 5.0 |
| 11 | 4.0 | 130.0 | 82.0 | 25.0 | 150.0 | 36.1 | 0.7 | 40.0 | 1.0 | 5.0 |
| 12 | 0.0 | 110.0 | 70.0 | 10.0 | 0.0 | 24.5 | 0.2 | 22.0 | 0.0 | 5.0 |
| 13 | 13.0 | 126.0 | 90.0 | 0.0 | 0.0 | 43.4 | 0.583 | 42.0 | 1.0 | nan |
| 14 | 1.0 | 119.0 | 44.0 | 47.0 | 63.0 | 35.5 | 0.28 | 25.0 | 0.0 | nan |
| 15 | 1.0 | 181.0 | 78.0 | 42.0 | 293.0 | 40.0 | 1.258 | 22.0 | 1.0 | nan |
| 16 | 3.0 | 78.0 | 70.0 | 0.0 | 0.0 | 32.5 | 0.27 | 39.0 | 0.0 | nan |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.0 | 81.0 | 86.0 | 16.0 | 66.0 | 27.5 | 0.306 | 22.0 | 0.0 | 6.0 |
| 2 | 5.0 | 137.0 | 108.0 | 0.0 | 0.0 | 48.8 | 0.227 | 37.0 | 1.0 | 6.0 |
| 3 | 2.0 | 120.0 | 54.0 | 0.0 | 0.0 | 26.8 | 0.455 | 27.0 | 0.0 | 6.0 |
| 4 | 2.0 | 99.0 | 0.0 | 0.0 | 0.0 | 22.2 | 0.108 | 23.0 | 0.0 | 6.0 |
| 5 | 1.0 | 113.0 | 64.0 | 35.0 | 0.0 | 33.6 | 0.543 | 21.0 | 1.0 | 6.0 |
| 6 | 3.0 | 125.0 | 58.0 | 0.0 | 0.0 | 31.6 | 0.151 | 24.0 | 0.0 | 6.0 |
| 7 | 4.0 | 110.0 | 70.0 | 25.0 | 0.0 | 32.5 | 0.45 | 30.0 | 1.0 | 6.0 |
| 8 | 3.0 | 95.0 | 64.0 | 30.0 | 0.0 | 28.7 | 0.3 | 27.0 | 0.0 | 6.0 |
| 9 | 5.0 | 145.0 | 80.0 | 20.0 | 150.0 | 36.2 | 0.6 | 38.0 | 1.0 | 6.0 |
| 10 | 2.0 | 130.0 | 72.0 | 0.0 | 0.0 | 29.5 | 0.25 | 22.0 | 0.0 | 6.0 |
| 11 | 6.0 | 160.0 | 85.0 | 35.0 | 200.0 | 40.1 | 0.7 | 45.0 | 1.0 | 6.0 |
| 12 | 1.0 | 88.0 | 60.0 | 15.0 | 0.0 | 24.5 | 0.15 | 21.0 | 0.0 | 6.0 |
| 13 | 0.0 | 102.0 | 52.0 | 0.0 | 0.0 | 25.1 | 0.078 | 21.0 | 0.0 | nan |
| 14 | 4.0 | 76.0 | 62.0 | 0.0 | 0.0 | 34.0 | 0.391 | 25.0 | 0.0 | nan |
| 15 | 1.0 | 103.0 | 30.0 | 38.0 | 83.0 | 43.3 | 0.183 | 33.0 | 0.0 | nan |
| 16 | 1.0 | 124.0 | 74.0 | 36.0 | 0.0 | 27.8 | 0.1 | 30.0 | 0.0 | nan |

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.0 | 139.0 | 62.0 | 41.0 | 480.0 | 40.7 | 0.536 | 21.0 | 0.0 | 7.0 |
| 2 | 0.0 | 86.0 | 68.0 | 32.0 | 0.0 | 35.8 | 0.238 | 25.0 | 0.0 | 7.0 |
| 3 | 2.0 | 81.0 | 72.0 | 15.0 | 76.0 | 30.1 | 0.547 | 25.0 | 0.0 | 7.0 |
| 4 | 8.0 | 154.0 | 78.0 | 32.0 | 0.0 | 32.4 | 0.443 | 45.0 | 1.0 | 7.0 |
| 5 | 3.0 | 150.0 | 76.0 | 0.0 | 0.0 | 21.0 | 0.207 | 37.0 | 0.0 | 7.0 |
| 6 | 12.0 | 88.0 | 74.0 | 40.0 | 54.0 | 35.3 | 0.378 | 48.0 | 0.0 | 7.0 |
| 7 | 3.0 | 145.0 | 70.0 | 35.0 | 120.0 | 32.5 | 0.5 | 29.0 | 1.0 | 7.0 |
| 8 | 1.0 | 95.0 | 60.0 | 20.0 | 0.0 | 26.5 | 0.3 | 22.0 | 0.0 | 7.0 |
| 9 | 5.0 | 130.0 | 80.0 | 25.0 | 85.0 | 34.0 | 0.4 | 34.0 | 1.0 | 7.0 |
| 10 | 2.0 | 110.0 | 68.0 | 30.0 | 0.0 | 28.0 | 0.2 | 24.0 | 0.0 | 7.0 |
| 11 | 4.0 | 160.0 | 90.0 | 40.0 | 150.0 | 36.5 | 0.6 | 38.0 | 1.0 | 7.0 |
| 12 | 0.0 | 85.0 | 65.0 | 15.0 | 0.0 | 24.0 | 0.15 | 21.0 | 0.0 | 7.0 |
| 13 | 2.0 | 85.0 | 65.0 | 0.0 | 0.0 | 39.6 | 0.93 | 27.0 | 0.0 | nan |
| 14 | 2.0 | 142.0 | 82.0 | 18.0 | 64.0 | 24.7 | 0.761 | 21.0 | 0.0 | nan |
| 15 | 9.0 | 102.0 | 76.0 | 37.0 | 0.0 | 32.9 | 0.665 | 46.0 | 1.0 | nan |
| 16 | 4.0 | 117.0 | 64.0 | 27.0 | 120.0 | 33.2 | 0.23 | 24.0 | 0.0 | nan |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.0 | 126.0 | 88.0 | 41.0 | 235.0 | 39.3 | 0.704 | 27.0 | 0.0 | 8.0 |
| 2 | 5.0 | 162.0 | 104.0 | 0.0 | 0.0 | 37.7 | 0.151 | 52.0 | 1.0 | 8.0 |
| 3 | 7.0 | 81.0 | 78.0 | 40.0 | 48.0 | 46.7 | 0.261 | 42.0 | 0.0 | 8.0 |
| 4 | 8.0 | 108.0 | 70.0 | 0.0 | 0.0 | 30.5 | 0.955 | 33.0 | 1.0 | 8.0 |
| 5 | 11.0 | 155.0 | 76.0 | 28.0 | 150.0 | 33.3 | 1.353 | 51.0 | 1.0 | 8.0 |
| 6 | 1.0 | 181.0 | 78.0 | 42.0 | 293.0 | 40.0 | 1.258 | 22.0 | 1.0 | 8.0 |
| 7 | 2.0 | 110.0 | 72.0 | 20.0 | 85.0 | 28.5 | 0.35 | 26.0 | 0.0 | 8.0 |
| 8 | 4.0 | 145.0 | 76.0 | 24.0 | 150.0 | 32.1 | 0.45 | 34.0 | 1.0 | 8.0 |
| 9 | 1.0 | 95.0 | 60.0 | 15.0 | 0.0 | 24.0 | 0.2 | 22.0 | 0.0 | 8.0 |
| 10 | 3.0 | 130.0 | 80.0 | 30.0 | 120.0 | 30.0 | 0.3 | 29.0 | 0.0 | 8.0 |
| 11 | 5.0 | 160.0 | 85.0 | 35.0 | 200.0 | 34.5 | 0.6 | 40.0 | 1.0 | 8.0 |
| 12 | 0.0 | 90.0 | 70.0 | 0.0 | 0.0 | 26.5 | 0.15 | 21.0 | 0.0 | 8.0 |
| 13 | 1.0 | 119.0 | 44.0 | 47.0 | 63.0 | 35.5 | 0.28 | 25.0 | 0.0 | nan |
| 14 | 8.0 | 196.0 | 76.0 | 29.0 | 280.0 | 37.5 | 0.605 | 57.0 | 1.0 | nan |
| 15 | 1.0 | 118.0 | 58.0 | 36.0 | 94.0 | 33.3 | 0.261 | 23.0 | 0.0 | nan |
| 16 | 1.0 | 116.0 | 70.0 | 28.0 | 0.0 | 27.4 | 0.204 | 21.0 | 0.0 | nan |

# TabuLLM: AI Benchmark Report

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4.0 | 96.0 | 56.0 | 17.0 | 49.0 | 20.8 | 0.34 | 26.0 | 0.0 | 9.0 |
| 2 | 0.0 | 102.0 | 52.0 | 0.0 | 0.0 | 25.1 | 0.078 | 21.0 | 0.0 | 9.0 |
| 3 | 7.0 | 124.0 | 70.0 | 33.0 | 215.0 | 25.5 | 0.161 | 37.0 | 0.0 | 9.0 |
| 4 | 7.0 | 150.0 | 66.0 | 42.0 | 342.0 | 34.7 | 0.718 | 42.0 | 0.0 | 9.0 |
| 5 | 0.0 | 108.0 | 68.0 | 20.0 | 0.0 | 27.3 | 0.787 | 32.0 | 0.0 | 9.0 |
| 6 | 1.0 | 97.0 | 70.0 | 40.0 | 0.0 | 38.1 | 0.218 | 30.0 | 0.0 | 9.0 |
| 7 | 3.0 | 85.0 | 66.0 | 20.0 | 0.0 | 28.4 | 0.237 | 29.0 | 0.0 | 9.0 |
| 8 | 5.0 | 140.0 | 80.0 | 30.0 | 120.0 | 36.5 | 0.627 | 45.0 | 1.0 | 9.0 |
| 9 | 2.0 | 95.0 | 70.0 | 25.0 | 0.0 | 31.2 | 0.456 | 24.0 | 0.0 | 9.0 |
| 10 | 1.0 | 110.0 | 68.0 | 0.0 | 0.0 | 27.5 | 0.289 | 22.0 | 0.0 | 9.0 |
| 11 | 4.0 | 130.0 | 75.0 | 22.0 | 90.0 | 34.1 | 0.512 | 38.0 | 1.0 | 9.0 |
| 12 | 0.0 | 100.0 | 60.0 | 0.0 | 0.0 | 26.3 | 0.215 | 21.0 | 0.0 | 9.0 |
| 13 | 8.0 | 65.0 | 72.0 | 23.0 | 0.0 | 32.0 | 0.6 | 42.0 | 0.0 | nan |
| 14 | 10.0 | 75.0 | 82.0 | 0.0 | 0.0 | 33.3 | 0.263 | 38.0 | 0.0 | nan |
| 15 | 2.0 | 146.0 | 70.0 | 38.0 | 360.0 | 28.0 | 0.337 | 29.0 | 1.0 | nan |
| 16 | 5.0 | 126.0 | 78.0 | 27.0 | 22.0 | 29.6 | 0.439 | 40.0 | 0.0 | nan |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunc | Age | Outcome | _fold_id |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.0 | 91.0 | 54.0 | 25.0 | 100.0 | 25.2 | 0.234 | 23.0 | 0.0 | 10.0 |
| 2 | 7.0 | 196.0 | 90.0 | 0.0 | 0.0 | 39.8 | 0.451 | 41.0 | 1.0 | 10.0 |
| 3 | 2.0 | 112.0 | 66.0 | 22.0 | 0.0 | 25.0 | 0.307 | 24.0 | 0.0 | 10.0 |
| 4 | 11.0 | 103.0 | 68.0 | 40.0 | 0.0 | 46.2 | 0.126 | 42.0 | 0.0 | 10.0 |
| 5 | 2.0 | 95.0 | 54.0 | 14.0 | 88.0 | 26.1 | 0.748 | 22.0 | 0.0 | 10.0 |
| 6 | 1.0 | 114.0 | 66.0 | 36.0 | 200.0 | 38.1 | 0.289 | 21.0 | 0.0 | 10.0 |
| 7 | 3.0 | 120.0 | 70.0 | 25.0 | 150.0 | 32.5 | 0.45 | 28.0 | 1.0 | 10.0 |
| 8 | 1.0 | 95.0 | 60.0 | 20.0 | 0.0 | 24.5 | 0.3 | 22.0 | 0.0 | 10.0 |
| 9 | 5.0 | 140.0 | 80.0 | 30.0 | 200.0 | 36.0 | 0.55 | 35.0 | 1.0 | 10.0 |
| 10 | 0.0 | 85.0 | 50.0 | 15.0 | 0.0 | 22.0 | 0.2 | 21.0 | 0.0 | 10.0 |
| 11 | 4.0 | 130.0 | 75.0 | 28.0 | 180.0 | 34.5 | 0.4 | 30.0 | 1.0 | 10.0 |
| 12 | 2.0 | 110.0 | 65.0 | 22.0 | 0.0 | 27.5 | 0.32 | 26.0 | 0.0 | 10.0 |
| 13 | 3.0 | 180.0 | 64.0 | 25.0 | 70.0 | 34.0 | 0.271 | 26.0 | 0.0 | nan |
| 14 | 3.0 | 99.0 | 80.0 | 11.0 | 64.0 | 19.3 | 0.284 | 30.0 | 0.0 | nan |
| 15 | 4.0 | 148.0 | 60.0 | 27.0 | 318.0 | 30.9 | 0.15 | 29.0 | 1.0 | nan |
| 16 | 7.0 | 136.0 | 90.0 | 0.0 | 0.0 | 29.9 | 0.21 | 50.0 | 0.0 | nan |